

MR KLEAN LAUNDRY: FROM DATA TO DECISIONS

Business Intelligence, Exploratory Data Analysis & Predictive Analytics Using R

Project Report

1. Introduction

Mr Klean Laundry accumulated customer, transaction, service, payment, delivery and operating-cost records. This project transforms the integrated business data into descriptive, diagnostic and predictive insights using R Programming. The overall workflow is structured as Data -> Pre-processing -> EDA -> Feature Selection -> Modelling -> Insights -> Decisions.

2. Project Objectives

Design and integrate a realistic laundry-business dataset.

Prepare the data for reliable analysis and modelling.

Use R for exploratory analysis, statistics and visualization.

Answer seven management questions covering customers, states, services, delivery, costs and monthly performance.

Identify important predictors of transaction-level NetProfit.

Build and compare six predictive models and select the best-performing model.

Translate analytical findings into practical recommendations.

3. Data Description

The final integrated dataset contains 1,000 records and 32 variables.

Variable classification

18 numerical variables: Age, QuantityOfClothes, WeightKg, Revenue, PaymentAmount, operating-cost variables, TotalExpense, NetProfit and related financial measures.

8 categorical variables: Gender, State, CustomerType, ServiceName, LoadType, DeliveryType, PaymentMethod and PaymentStatus.

4 date variables: OrderDate, PickupDate, DeliveryDate and PaymentDate.

2 identifier variables: OrderID and CustomerID.

Target variable for predictive modelling: NetProfit.

4. Source and Design of the Data

The dataset is a realistic synthetic business dataset created for the Mr Klean Laundry project. The relational structure was designed in MySQL and covers customers, orders, services, employees, machines, assignments, payments and expenses. Synthetic records were generated with Mockaroo, then imported into MySQL for validation, cleaning, transformation and integration. The resulting dataset was exported as a CSV file for analysis in R.

5. Data Pre-processing

Duplicate check: examine the dataset for repeated records before analysis.

Missing-value check: identify missing observations and determine appropriate treatment.

Outlier detection: inspect numerical variables for unusually high or low observations.

Data transformation: convert variables to appropriate data types and prepare categorical variables for modelling.

Scaling/standardization: apply where required by the modelling technique.

Target preparation: define NetProfit as the response variable.

Data-quality reporting note: The supplied presentation does not report numeric counts for duplicates, missing values or outliers. Therefore, this revised report documents the required preprocessing checks without inventing results that are not present in the supplied material.

Outlier Interpretation - Weight (Kg)

The WeightKg boxplot shows most observations concentrated around 4-11 kg. The upper whisker extends to approximately 22 kg, with several observations above this level reaching about 32 kg. These observations were treated as statistical outliers. They were retained because unusually heavy loads can represent valid high-volume laundry orders rather than data-entry errors.

Fig. 13. Boxplot of Weight (Kg): Outlier Detection

6. Exploratory Data Analysis

EDA was used to understand the structure and behaviour of the business data before predictive modelling. The analysis includes univariate, bivariate and multivariate analysis using bar charts, trend charts and correlation analysis.

7. Customer Category Analysis

Which customer categories generate the highest Revenue and NetProfit?

Customer Category was compared with Revenue and NetProfit.

The multivariate bar chart compares two numerical financial measures across one categorical variable.

Regular customers are the strongest contributor in the documented chart.

Business use: customer retention, acquisition and segment prioritization.

Fig. 1. Revenue and Net Profit by Customer Category

8. State Contribution Analysis

Do customers from different states contribute differently to revenue and profitability?

Revenue and NetProfit were compared across customer states.

The horizontal format improves readability across many states.

Business use: identify stronger markets and areas requiring targeted marketing or service improvement.

Fig. 2. Revenue and Net Profit by State

9. Service Revenue and Profitability

Which laundry services generate the highest revenue and profit?

Laundry recorded the highest total revenue and net profit.

Revenue: \$3,862,076.

Net Profit: \$1,959,697.

Business use: protect capacity and service quality in strong services while reviewing weaker services.

Fig. 3. Revenue and Net Profit by Laundry Service

10. Quantity, Weight and Revenue

Is there a relationship between quantity, weight and revenue?

Quantity <-> Weight: $r = 0.753$.

Quantity <-> Revenue: $r = 0.009$.

Weight <-> Revenue: $r = 0.024$.

Quantity and weight are strongly related to each other, but neither has a meaningful linear relationship with revenue.

Business use: consider service type, pricing, customer and delivery factors.

Fig. 4. Relationship Between Quantity, Weight and Revenue

11. Delivery Option Analysis

Which delivery option contributes most to revenue and profitability?

Customer Pickup is the strongest delivery option.

Revenue: \$4.55M.

Net Profit: \$2.51M.

Home Delivery is second; Express contributes the least.

Business use: maintain strong pickup capacity and review the economics of other options.

Fig. 5. Revenue and Net Profit by Delivery Option

12. Operational Cost Analysis

Which operating costs contribute most to overall expenses?

Labour Cost: \$1.63M.

Tax: \$587,944.

Electricity: \$390,188.

Maintenance and Detergent are also significant.

Business use: focus cost-control efforts on the largest cost drivers.

Fig. 6. Operational Cost Components

13. Monthly Demand Analysis

Are there monthly or seasonal patterns in demand, revenue and profitability?

Monthly demand fluctuates substantially.

High-demand months generally produce higher revenue and profit.

No consistent recurring seasonal pattern was established in the documented period.

Business use: flexible capacity planning and close monthly monitoring.

Fig. 7. Monthly Customer Demand

14. Monthly Revenue Trend

Revenue shows irregular peaks and troughs across the documented months. Revenue and NetProfit generally move in the same direction, although the monthly series is not smooth.

Fig. 8. Monthly Revenue Trend

The corresponding monthly NetProfit trend is documented in the original presentation; it follows a similar irregular pattern.

15. Feature Selection and Correlation Analysis

Feature selection is included before predictive modelling to identify informative predictors and reduce unnecessary model complexity. Pearson correlation was used as an exploratory screening step.

The documented relationships are:

Quantity <-> Weight: $r = 0.753$ - strong positive relationship.

Quantity <-> Revenue: $r = 0.009$ - virtually no linear relationship.

Weight <-> Revenue: $r = 0.024$ - virtually no linear relationship.

Fig. 9. Correlation Matrix of the Documented Numeric Relationships

Encoding note: Categorical variables must be encoded numerically for modelling. The supplied presentation does not contain the original full 32-variable encoded correlation matrix, so this report presents only the documented numeric correlations rather than fabricating a complete matrix.

16. Feature Importance for NetProfit

Model-based feature importance provides a second screening mechanism for identifying variables that contribute strongly to the selected predictive model.

LabourCost has the highest recorded relative importance (100).

Tax follows at 58.48.

MaintenanceCost and ElectricityCost are the next major contributors.

The result is consistent with the descriptive cost analysis.

Fig. 10. PLS Variable Importance

17. Step-by-Step Predictive Modelling Process

10. Test and Compare - Evaluate RMSE, R2 and MAE, rank the models, and inspect Actual-vs-Predicted plots.

9. Generate Predictions - Use each trained model to predict NetProfit for the 200-observation test set.

8. Tune the Models - PLS was tuned using tuneLength = 20 within the cross-validation framework.

7. Build the Six Models - Random Forest, Bagging, GBM, SVM, PLS and kNN.

6. Apply 10-Fold Cross-Validation - Use the same resampling conditions for all candidate models so their performance can be compared fairly.

5. Set the Seed - set.seed(123) was used to make the modelling results reproducible.

4. Split the Data - Use 80% for training (800 observations) and 20% for testing (200 observations).

3. Prepare the Modelling Data - Encode categorical variables and apply scaling/standardization where required by the modelling technique.

2. Feature Importance - Use model-based feature importance to identify the strongest contributors to NetProfit. LabourCost recorded the highest importance (100), followed by Tax (58.48).

1. Feature Selection & Correlation - Identify informative predictors; use Pearson correlation as an exploratory screening step and assess multicollinearity with VIF.

18. Model Validation and Testing

The modelling workflow used an 80/20 train/test split, with 800 observations for training and 200 observations reserved for final test-set evaluation. All candidate models were trained using the same 10-fold cross-validation control so that their performance could be compared under consistent resampling conditions.

Feature screening included Pearson correlation analysis, numerical preparation of categorical variables, and VIF assessment for multicollinearity. The PLS model was tuned using tuneLength = 20. After training, predictions were generated for the test set and evaluated using RMSE, R2 and MAE. Model ranking was based on these metrics, and Actual-versus-Predicted plots were used as visual diagnostics. The recorded PLS R2 of 1.0000 is exceptionally high, so a target-leakage check is required before production use.

The recorded PLS R2 of 1.0000 is exceptionally high. A target-leakage check is therefore required before treating the result as a production-ready model, particularly because NetProfit is a derived financial outcome.

19. Predictive Modelling - Six Models Compared

The target variable is NetProfit. Six predictive models were compared: Partial Least Squares (PLS), Gradient Boosting Machine (GBM), Random Forest, Support Vector Machine (SVM), k-Nearest Neighbours (kNN), and Bagging.

Fig. 11. Model Comparison by R-squared

Fig. 12. Model Comparison by MAE (Lower is Better)

Six-Model Performance Comparison

Interpretation: lower RMSE and MAE indicate smaller prediction errors, while higher R2 indicates greater explained variation. The recorded ranking is PLS, GBM, Random Forest, SVM, kNN and Bagging.

21. Actual vs Predicted Model Diagnostics

Fig. 14. Bagging Regression: Actual vs Predicted

Predictions follow the overall upward relationship but show considerable dispersion around the reference line, consistent with the highest recorded prediction errors.

Fig. 15. Random Forest Regression: Actual vs Predicted

Predictions are tightly clustered around the reference line, indicating strong agreement between actual and predicted NetProfit.

Fig. 16. GBM Regression: Actual vs Predicted

Points lie very close to the reference line across most of the range, indicating a strong predictive fit.

Fig. 17. SVM Regression: Actual vs Predicted

The model captures the overall relationship but shows more dispersion around the reference line, particularly at lower NetProfit values.

Fig. 18. PLS Regression: Actual vs Predicted Net Profit

The points almost completely overlap the reference line, visually supporting the recorded PLS metrics. The unusually perfect fit requires a target-leakage check.

Fig. 19. kNN Regression: Actual vs Predicted Net Profit

The model captures the upward trend but shows wider scatter around the reference line, indicating weaker predictive precision than the strongest models.

27. Selected Model - PLS

Recorded ranking: PLS (Partial Least Squares) -> GBM (Gradient Boosting Machine) -> Random Forest -> SVM (Support Vector Machine) -> kNN (k-Nearest Neighbours) -> Bagging (Bootstrap Aggregating).

RMSE = 0.0336.

R-squared = 1.0000.

MAE = 0.0280.

PLS was selected as the project model based on the recorded comparison. Its R2 = 1.0000 and very small RMSE/MAE should, however, be treated with caution until target leakage has been ruled out.

Validation caution: An R-squared of 1.0000 is exceptionally high. Before presenting this as a final portfolio result, verify that no target-derived variables or direct components of NetProfit were included among the predictors, because that could cause target leakage.

28. Overall Findings

Regular customers are a major contributor to Revenue and NetProfit in the documented customer analysis.

Laundry is the strongest-performing service in the documented service analysis.

Customer Pickup is the strongest delivery option.

Labour Cost is the largest operational expense.

Quantity and Weight are strongly related, but neither is meaningfully correlated with Revenue.

Monthly performance fluctuates substantially without a clearly established recurring seasonal pattern.

LabourCost has the highest recorded PLS variable importance.

PLS ranks first among the six recorded predictive models.

29. Recommendations

Improve staff scheduling, workload distribution and labour productivity.

Monitor Labour, Electricity, Maintenance and Detergent costs closely.

Maintain strong Customer Pickup capacity and service quality.

Use customer and service profitability analysis to guide retention, marketing and capacity decisions.

Do not rely on quantity or weight alone for pricing or revenue expectations.

Use monthly monitoring to respond early to weak and high-demand periods.

Continuously validate and retrain the predictive model as new business data becomes available.

30. Conclusion

The project transformed integrated Mr Klean Laundry data into actionable business intelligence using R. The analysis combined preprocessing, exploratory data analysis, correlation analysis, feature importance and predictive modelling. The major management priorities are labour efficiency, cost control, customer profitability, service performance and delivery operations. The documented modelling comparison selected PLS as the best-performing model. Overall, the project establishes a repeatable framework: structured data -> analysis -> insight -> prediction -> action.

24. Final Management Message

The value of the project is not only the charts or the machine-learning model. The value lies in establishing a repeatable process through which structured business data can be converted into insight, prediction and action. Mr Klean Laundry can use this framework to improve operational efficiency, control costs and strengthen profitability.

Model | RMSE ↓ | R2 ↑ | MAE ↓ | Performance

PLS (Partial Least Squares) | 0.0336 | 1.0000 | 0.0280 | Excellent

GBM (Gradient Boosting Machine) | 125.0357 | 0.9907474 | 90.0455 | Very Good

Random Forest | 300.1462 | 0.9827295 | 215.2852 | Very Good

SVM (Support Vector Machine) | 450.6877 | 0.9622112 | 270.0380 | Good

kNN (k-Nearest Neighbours) | 405.5169 | 0.9515488 | 334.1182 | Good

Bagging (Bootstrap Aggregating) | 535.7497 | 0.9464361 | 421.0446 | Fair